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Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Customer Journey Map

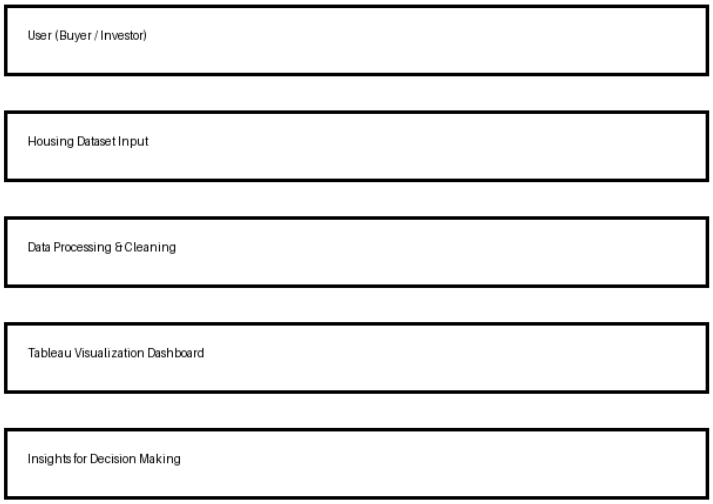
Stage	User Actions	User Thoughts	Pain Points	Opportunities
Search	Searches housing listings	Which area is affordable?	Too many listings	Provide filtered dashboards
Compare	Compares prices & features	Which house gives best value?	Data scattered	Visual comparisons
Analyze	Looks at charts & reports	Are prices increasing?	Hard to interpret data	Interactive Tableau charts
Decide	Shortlists properties	Is this a good investment?	Uncertain trends	Clear trend insights

Brainstorming Ideas

- Create interactive Tableau dashboards for housing prices.
- Analyze relationship between house size and price.
- Compare prices across different locations.
- Identify features that influence property prices.
- Show yearly housing market trends.
- Provide filters for bedrooms, location, and amenities.

Data Flow Diagram (DFD)

The following diagram shows how housing data moves through the system to generate insights.



1. Solution Kit

The Solution Kit explains how the project will solve the problem using tools and methods.

Tools Used

- Tableau – Data visualization and dashboard creation
- Excel / CSV Dataset – Housing market dataset
- Data Cleaning Techniques – Removing missing or inconsistent values
- Data Analysis – Identifying relationships between price and features

Steps in the Solution

1. Data Collection
    - Collect housing dataset containing:
        - Sale price
        - Location
        - Number of bedrooms
        - Square footage
        - Year built
        - Property condition
2. Data Cleaning
    - Handle missing values
    - Remove duplicate records
    - Standardize column names

3. Data Processing
    - Categorize houses by price range
    - Group properties by location
    - Calculate average prices
4. Visualization
    - Create interactive Tableau dashboards
    - Use charts to show price trends and feature impact
5. Insight Generation
    - Identify factors affecting housing prices
    - Provide insights for buyers and investors

2. Requirement Analysis

Requirement analysis identifies what the system needs to work properly.

Functional Requirements

The system should:

1. Import housing dataset (CSV / Excel)
2. Clean and process housing data
3. Analyze sale prices and property features
4. Generate visualizations using Tableau
5. Allow users to filter data by:
    - Location
    - Price range
    - Number of bedrooms
6. Display trends and comparisons

Non-Functional Requirements

Requirement	Description
Usability	Dashboard should be simple and easy to understand
Performance	Data should load quickly
Accuracy	Visualizations must reflect correct data
Reliability	System should produce consistent insights
Scalability	Should handle larger housing datasets

3. Tableau Dashboard Design

The Tableau dashboard will visualize housing market insights.

Dashboard Components

1. Price Trend Chart

- Line chart
- Shows housing price trends over time

2. Price by Location

- Map or bar chart
- Compares housing prices across locations

3. Feature Impact Analysis

- Scatter plot
- Shows relationship between:
    - house size
    - number of bedrooms
    - sale price

4. Property Distribution

- Pie chart / bar chart
- Distribution of houses by:
    - property type
    - condition
    - bedrooms

5. Filters (Interactive)

Users can filter data by:

- Location
- Price range
- Bedrooms
- Year built

Proposed Solution

This project develops an interactive Tableau dashboard that visualizes housing market trends based on sale prices and property features. The system helps buyers, investors, and analysts quickly understand market behavior, compare property attributes, and make better real estate decisions.